

# Lecture Notes, Fast

A practical LaTeX + VimTeX desk reference

## The 10-second loop

Type notes with snippets and the short formatting triggers. Let the 650 ms save debounce do its work. When you want a checkpoint: **Space la** saves and compiles; **Space lv** opens the PDF; **Space le** shows errors.

## Your key language

| Key                 | Action                    |
|---------------------|---------------------------|
| <b>Space</b>        | leader key for commands   |
| <b>jk</b>           | leave Insert mode quickly |
| <b>Space w</b>      | save current file         |
| <b>Space q</b>      | quit window               |
| <b>Ctrl-h/j/k/l</b> | move between splits       |
| <b>Space zw</b>     | distraction-free writing  |

## Compile and inspect

| Keys            | Use                              |
|-----------------|----------------------------------|
| <b>Space ll</b> | start/refresh continuous compile |
| <b>Space la</b> | save, then compile               |
| <b>Space lv</b> | view PDF in Skim                 |
| <b>Space le</b> | inspect compiler errors          |
| <b>Space lt</b> | toggle table of contents         |
| <b>Space lk</b> | stop compiler                    |
| <b>Space lm</b> | one-shot compile                 |
| <b>Space lo</b> | show compiler output             |
| <b>Space lc</b> | clean auxiliary files            |
| <b>Space li</b> | show VimTeX project info         |
| <b>Space ts</b> | toggle spell checking            |

Quickfix does not open automatically.

Use **Space le** only when you want diagnostics, so your cursor and page stay stable while lecturing.

## Find things without leaving the file

|                 |                                |
|-----------------|--------------------------------|
| <b>Space ff</b> | find a file                    |
| <b>Space fg</b> | search text across the project |
| <b>Space fs</b> | jump through document symbols  |
| <b>Space fb</b> | choose an open buffer          |
| <b>Space e</b>  | toggle the file tree           |

## The fast lecture pattern

1. Open or create a ‘.tex’ file. A new file receives your template automatically.
2. Type a trigger such as **defn**, **thm**, **proof**, **al**, or **eq**; expand it with your completion key (usually **Tab**).
3. Use **;c** for a fresh bold Claim paragraph, **;p** for Proof, and **;r** for Remark.
4. Press **Space la** at a natural pause. If something breaks, press **Space le**.

## Formatting while you type

### One-line emphasis

In Insert mode:

|           |                                                              |
|-----------|--------------------------------------------------------------|
| <b>;b</b> | inserts <code>\textbf{}</code> and places the cursor inside. |
| <b>;i</b> | inserts <code>\emph{}</code> and places the cursor inside.   |
| <b>;c</b> | starts a new paragraph labeled <b>Claim</b> .                |
| <b>;p</b> | starts a new paragraph labeled <b>Proof</b> .                |
| <b>;r</b> | starts a new paragraph labeled <b>Remark</b> .               |
| <b>\(</b> | inserts <code>\(</code> with the cursor inside.              |
| <b>\[</b> | inserts <code>\[</code> with the cursor inside.              |

## LuaSnip triggers already installed

### Statements and proof structure

|           |                                                               |
|-----------|---------------------------------------------------------------|
| prb       | problem plus solution                                         |
| prob      | numbered problem                                              |
| hw        | <code>\exercise{#.#}</code> header + solution (solutions.tex) |
| chb       | chapter banner (solutions.tex)                                |
| sol, ans  | solution or answer block                                      |
| defn      | definition                                                    |
| thm, lem, | theorem, lemma, proposition, corollary                        |
| prop, cor |                                                               |
| ex, rem,  | example, remark, proof                                        |
| proof     |                                                               |

### Display and layout

|            |                           |
|------------|---------------------------|
| al, eq     | align or equation display |
| cases, mat | cases or matrix           |
| enum, item | numbered or bulleted list |
| sec, ssec  | section or subsection     |

### Mathematical shorthand

|                 |                                                              |
|-----------------|--------------------------------------------------------------|
| rr, cc, qq, zz, | $\mathbb{R}, \mathbb{C}, \mathbb{Q}, \mathbb{Z}, \mathbb{N}$ |
| nn              |                                                              |
| abs, norm, ip,  | absolute value, norm, inner product,                         |
| set             | set                                                          |
| span, rank,     | common linear/algebra operators                              |
| ker, im         |                                                              |
| lim, int, pd,   | limits, integrals, partial/ordinary                          |
| dv              | derivatives                                                  |
| pset, cond, ev, | probability notation                                         |
| var, cov        |                                                              |

## Good source habits

- Each course folder keeps one growing `solutions.tex` with a banner per textbook chapter. Jump to the chapter (Toc: `Space 1t`), type `hw`, press `Tab`, and write.
- Put one mathematical claim per paragraph. Start it with `;c`.
- Use `align*` for multiple equalities; put ‘&’ before the alignment point and ‘  
’ between lines.
- Use `\text{...}` inside math for words:  $x > 0$  if  $x \neq 0$ .
- Label important results: `\label{thm:name}`; reference them with `\cref{thm:name}`.
- Keep scratch explanations in `\textbf{Claim.}`, `{Proof.}`, or `{Idea.}` rather than burying them in comments.

### Useful emergency commands

|                               |                               |
|-------------------------------|-------------------------------|
| <code>:VimtexCompile</code>   | start/refresh compile         |
| <code>:VimtexView</code>      | open the PDF                  |
| <code>:VimtexErrors</code>    | inspect errors                |
| <code>:VimtexTocToggle</code> | show/hide outline             |
| <code>:VimtexStop</code>      | stop compilation              |
| <code>:VimtexInfo</code>      | inspect project configuration |

### Remember

If a shortcut is forgotten: press `Space` and wait for which-key/help prompts, or use `:h vimtex`.